|INCH-POUND|
MIL-M-28787/433
AMENDMENT 1
17 June 1992

MILITARY SPECIFICATION

MODULES, STANDARD ELECTRONIC

MEMORY TIMING MODULE, KEY CODE JEQ

This amendment forms a part of Military Specification MIL-M-28787/433, dated 6 August 1991, and is approved for use by all Departments and Agencies of the Department of Defense.

PAGES 95 through 104

Append the attached pages to the end of the specification.

Custodians:
Army - ER
Navy - SH
Air Force - 85

Review activities:
Army - AT
Navy - AS, MC
Air Force - 13, 19
DLA - ES

Preparing activity:
Navy - NW

(Project 5963-0271)

AMSC N/A FSC 5963
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TABLE XVIII. Input and output capacitance test.

Test cond	Signal name	Pin no.	25 ± 5°C Initial limits Min	Max	Units	Notes
Inputs						
R1	CBUS A/WR D2	3	2	50	pF	1-4
R2	SOFT ERROR	4	8	125	pF	1-4
R3	CBUS A/WR D1	5	2	50	pF	1-4
R4	CBUS A/WR D0	6				
R5	WR DATA 15	8				
R6	WR DATA 14	9	2	50	pF	1-4
R7	HERE	11	6	100	pF	1-4
R8	HOLE DET 19	14	2	50	pF	1-4
R9	HOLE DET 18	15	4	75	pF	1-4
R10	HOLE DET 16	16	8	100	pF	1-4
R11	HOLE DET 17	17	6	75	pF	1-4
R12	SIZE 1	18	4	75	pF	1-4
R13	SIZE 0	19	4	75	pF	1-4
R14	TYPE 0	20	6	75	pF	1-4
R15	LATCH ST REG	21	6	75	pF	1-4
R16	MEM D BUS 5	22	2	50	pF	1-4
R17	MEM D BUS 2	23	2	50	pF	1-4
R18	MEM D BUS 4	24	4	50	pF	1-4
R19	MEM D BUS 7	26	2	50	pF	1-4
R20	MEM D BUS 11	29	2	50	pF	1-4

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TABLE XVIII. Input and output capacitance test - Continued.

Test cond	Signal name	Pin no.	25 ± 5°C Initial limits Min	Max	Units	Notes
R21	MEM D BUS 9	30	2	50	pF	1-4
R22	MEM D BUS 10	32				
R23	MEM D BUS 3	33				
R24	MEM D BUS 8	34	2	50	pF	1-4
R25	CYCLE INIT	35	2	75	pF	1-4
R26	MULTI BIT ER	36	6	75	pF	1-4
R27	INHIB ER COR	37	4	75	pF	1-4
R28	REFRESH CYC	41				
R29	PARITY ER	44	4	75	pF	1-4
R30	EN CBUS PARI	45	2	50	pF	1-4
R31	CBUS CYC ABT	48	2	50	pF	1-4
R32	NDRO LATCH	49	4	75	pF	1-4
R33	CLK A	53	4	75	pF	1-4
R34	CBUS D T/OUT	54	2	75	pF	1-4
R35	ECC ERROR	55	4	75	pF	1-4
R36	INIT RD EN	56	2	50	pF	1-4
R37	CB OR	57	4	100	pF	1-4
R38	READ TMG 1	58	10	125	pF	1-4
R39	LATCH RD D	59	5	150	pF	1-4
R40	STRAP BIT 1	66	2	75	pF	1-4

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TABLE XVIII. Input and output capacitance test - Continued.

Test cond	Signal name	Pin no.	25 ± 5°C Initial limits Min	Max	Units	Notes
R41	STRAP BIT 2	67	2	100	pF	1-4
R42	STRAP BIT 3	68	2	75	pF	1-4
R43	MID CYCLE	76	2	50	pF	1-4
R44	SLOW	77				
R45	PWR UP M/CLR	83	2	50	pF	1-4
R46	COM BUS CLK	84	2	100	pF	1-4
R47	25-75 NS	85	4	75	pF	1-4
R48	MMIO INHIB	91	6	125	pF	1-4
R49	SL DRAM LATH	92	4	75	pF	1-4
R50	SCRUB CYCLE	93	6	200	pF	1-4
R51	DELAY FCTN 2	94	2	150	pF	1-4
R52	DELAY FCTN 0	95	2	50	pF	1-4
R53	DELAY FCTN 1	96	2	50	pF	1-4
R54	DELAY FCTN 3	97	2	150	pF	1-4
R55	NON ARAY REF	98	2	125	pF	1-4
R56	LATCH ADRS	99	2	125	pF	1-4
Bidirectional						
R57	COM BUS BUSY	78	4	75	pF	1-4

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NOTES:

1. Input and output capacitance test tolerance table.

Condition	Value	Accuracy	Test condition
Capacitance meter frequency	1 MHz	± 100 KHz	R1-R57
Capacitance meter voltage	15 mV	± 1.5 mV	R1-R57
Capacitance measurement	NA	± 2.4 pF	R1
	NA	± 5.85 pF	R2
	NA	± 2.4 pF	R3-R6
	NA	± 4.7 pF	R7
	NA	± 2.4 pF	R8
	NA	± 3.55 pF	R9
	NA	± 4.6 pF	R10
	NA	± 3.45 pF	R11
	NA	± 3.55 pF	R12,R13
	NA	± 3.45 pF	R14,R15
	NA	± 2.4 pF	R16,R17
	NA	± 2.3 pF	R18
	NA	± 2.4 pF	R19-R24
	NA	± 3.65 pF	R25
	NA	± 3.45 pF	R26
	NA	± 3.55 pF	R27-R29
	NA	± 2.4 pF	R30,R31
	NA	± 3.55 pF	R32,R33
	NA	± 3.65 pF	R34
	NA	± 3.55 pF	R35
	NA	± 2.4 pF	R36
	NA	± 4.8 pF	R37
	NA	± 5.75 pF	R38
	NA	± 7.25 pF	R39
	NA	± 3.65 pF	R40
	NA	± 4.9 pF	R41
	NA	± 3.65 pF	R42
	NA	± 2.4 pF	R43-R45
	NA	± 4.9 pF	R46
	NA	± 3.55 pF	R47

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NOTES: - Continued.

Condition	Value	Accuracy	Test condition
Capacitance measurement	NA	± 5.95 pF	R48
	NA	± 3.55 pF	R49
	NA	± 9.7 pF	R50
	NA	± 7.4 pF	R51
	NA	± 2.4 pF	R52,R53
	NA	± 7.4 pF	R54
	NA	± 6.15 pF	R55,R56
	NA	± 3.55 pF	R57

2. The capacitance meter used in this test shall be item C05. The negative terminal shall be connected to the module ground and the positive terminal to the pin indicated in table XVIII.

3. The test circuit shall be as indicated on figure 12.

4. Abbreviations used are defined in 6.1.

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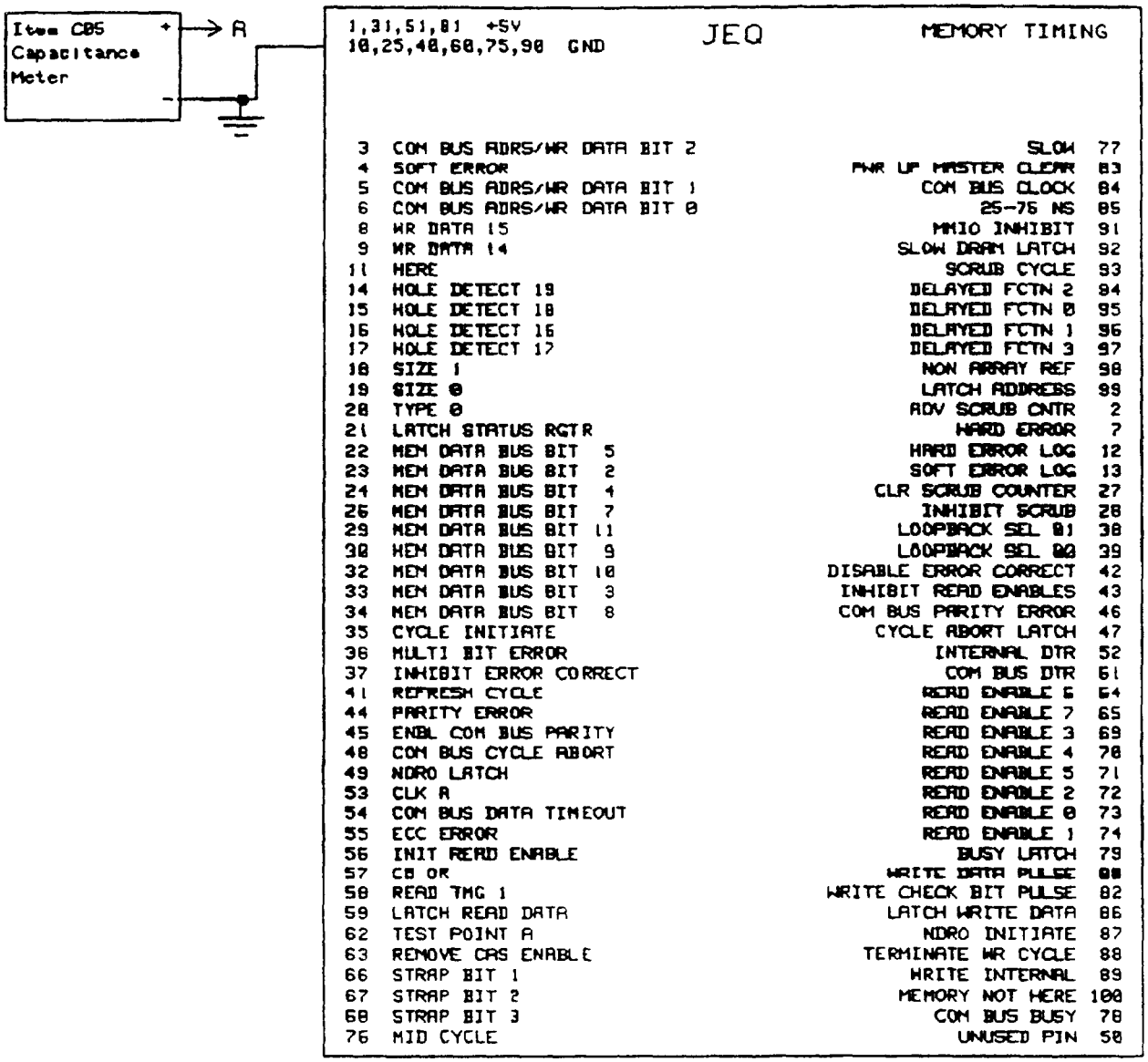


FIGURE 12. Input and output capacitance test circuit for table XVIII.

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TABLE XIX. Power supply transient amplitude test.

Test cond	Parameter	Pin no.	25 ± 5°C Initial limits Min	Max	Units	Notes
T1	PSTA	1	0	400	mV	1-4
T2	PSTA	51	0	400	mV	1-4

NOTES:

1. Power supply transient amplitude test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	5.0 V	± 250 mV	T1,T2
Power supply current limit	1.98 A	NA	T1,T2
Pattern voltage high, (VH) low, (VL)	 3.0 V 0.0 V	 ± 100 mV ± 100 mV	 T1,T2 T1,T2
Power supply transient amplitude measurement	NA	± 20 mV	T1,T2

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table to an external source. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

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NOTES: - Continued.

(b) Precondition the module under test and make the indicated measurement for the test condition shown in table XIX.

(c) Vcc pins 1 and 31 shall be shorted together, pins 51 and 81 shall be shorted together and the scope point, probe 'A', shall be connected to the pin indicated in table XIX.

(d) Using the oscilloscope, D06, inspect for the most negative spike as shown on figure 13 at the pattern line with the maximum power supply current.

(e) The maximum probe capacitance is 15 pF. There shall be no other capacitance present on Vcc to ground.

3. The test circuit shall be as indicated on figure 14.

4. Abbreviations used are defined in 6.1.

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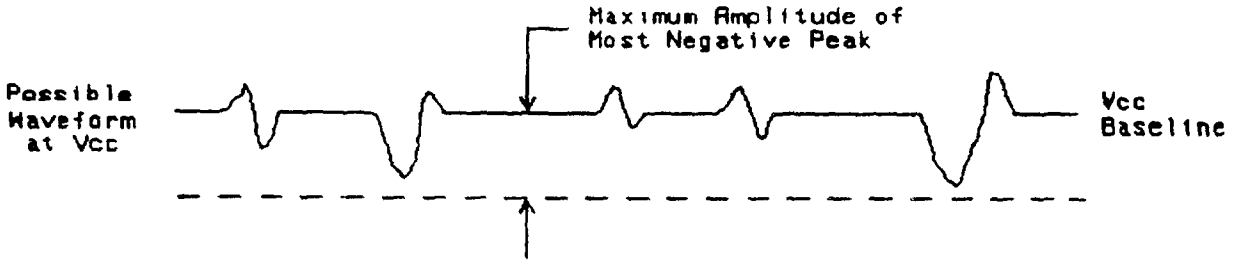
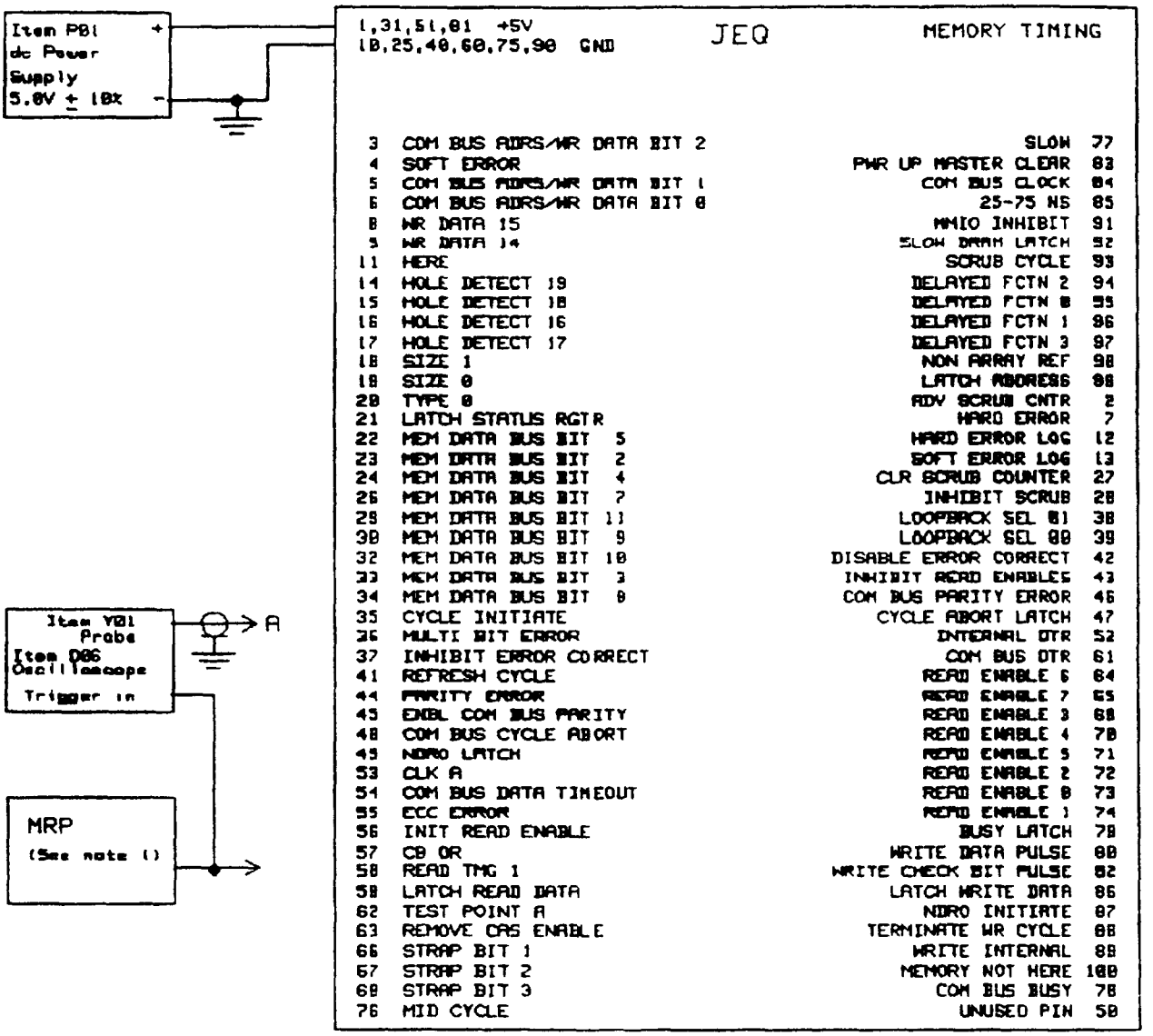


FIGURE 13. Power supply transient amplitude waveform for table XIX.

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NOTE:

1. MRP or emulator capable of reproducing system backpanel characteristics.

FIGURE 14. Power supply transient amplitude test circuit for table XIX.

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